A LISSE CONSTRUCTIBLE SHEAF ON A CONNECTED SCHEME OVER THE PRO-ETALE SITE IS ADIC LISSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification of the reductions through the comparison with the etale site and connectedness, the local description on a covering by affine weakly contractible schemes, and the comparison of the resulting finite modules through their reductions and completions.

Editorial note. The label lemma-connected-lisse names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-connected-lisse –start 5845 –end 5871 –local-id 3597 returned RESULT=OK (L3597 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Equivalence between locally constant sheaves on the etale and pro-etale sites. Same chapter, the lemma comparing locally constant sheaves of finite presentation on the two sites through the pullback. Supplies the locally constant sheaves G_n of Lambda/I^n-modules whose pullbacks are the reductions F/I^n F.
- Local constancy of a locally constant sheaf on a connected scheme. Etale Cohomology, the lemma that a locally constant sheaf of modules on a connected scheme is locally isomorphic to a constant sheaf on a finite module. Supplies the finite modules M_n with G_n locally isomorphic to the constant sheaf on M_n.
- Coverings by affine weakly contractible schemes. Same chapter, the lemma producing a covering by affine weakly contractible schemes. Supplies the covering {W_t -> X} on which the restrictions of F can be analysed.
- Description of lisse constructible sheaves on an affine weakly contractible scheme. Same chapter, the lemma identifying such a sheaf with the completion of a finite module. Supplies the finite completed modules N_t with F restricted to W_t isomorphic to the completion of N_t.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Isomorphism of finite modules with isomorphic completions. More on Algebra, the lemma that two finite modules over a Noetherian ring with isomorphic completions are isomorphic. Supplies the conclusion that the N_t are all isomorphic, which makes F adic lisse.

Source reference. Original lemma 09BF: sources/stacks-chow/tags.txt, source lines 11793–11793. Prerequisite pointer invoked at proof line(s) 5858; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 099Y: sources/stacks-proetale/pinned-original-source.tex, source lines 4338–4338. Prerequisite pointer invoked at proof line(s) 5854; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09BV: sources/stacks-proetale/pinned-original-source.tex, source lines 5783–5783. Prerequisite pointer invoked at proof line(s) 5864; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09BC: sources/stacks-chow/tags.txt, source lines 11790–11790. Prerequisite pointer invoked at proof line(s) 5869; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a connected scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. If $\mathcal{F}$ is a lisse constructible Λ-sheaf on $X_{pro\text{-}étale}$, then $\mathcal{F}$ is adic lisse.*

Proof. By Lemma 099Y we have $\mathcal{F}/I^n\mathcal{F} = \epsilon^{-1}\mathcal{G}_n$ for some locally constant sheaf $\mathcal{G}_n$ of Λ/I^n-modules. By Étale Cohomology, Lemma 09BF there exists a finite Λ/I^n-module M_n such that $\mathcal{G}_n$ is locally isomorphic to $\underline{M_n}$. Choose a covering $\{W_t \to X\}_{t\in T}$ with each W_t affine and weakly contractible. Then $\mathcal{F}|_{W_t}$ satisfies the assumptions of Lemma 09BV and hence $\mathcal{F}|_{W_t} \cong \underline{N_t}^\wedge$ for some finite $\Lambda^\wedge$-module N_t. Note that $N_t/I^nN_t \cong M_n$ for all t and n. Hence $N_t \cong N_{t'}$ for all $t, t' \in T$, see More on Algebra, Lemma 09BC. This proves that $\mathcal{F}$ is adic lisse. □